

Summary

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0.1 The Limit Laws:

Continuity If a function f is **continuous at** a , then

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Composition Limit Law If a function f is **continuous at** $b = \lim_{x \rightarrow a} g(x)$, then

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right)$$

For the limit laws below, assume $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ both exist. The following properties hold.

Sum Limit Law

$$\lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

Difference Limit Law

$$\lim_{x \rightarrow a} (f(x) - g(x)) = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

Product Limit Law

$$\lim_{x \rightarrow a} (f(x) \cdot g(x)) = \left(\lim_{x \rightarrow a} f(x)\right) \cdot \left(\lim_{x \rightarrow a} g(x)\right)$$

Quotient Limit Law

$$\lim_{x \rightarrow a} \left(\frac{f(x)}{g(x)}\right) = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}, \text{ provided } \lim_{x \rightarrow a} g(x) \neq 0$$

Limit laws also hold for one-sided limits.